# Toward a Standard Benchmark Format and Suite for Floating-Point Analysis




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# Incredible progress…

**Automatic Verification**
Fluctuat [SAS'13]
Rosa [POPL'14]
FPTaylor [FM'15]

**Optimization**
STOKE [PLDI'14]

**Improvement**
Salsa [FMICS'15]
Herbie [PLDI'15]

**Mechanized Proofs**
Wave equation [ITP'10]
Rounding error [NSV'16]

Rapid improvement in hard problems!

# Incredible progress…

We want our community
to keep progressing!

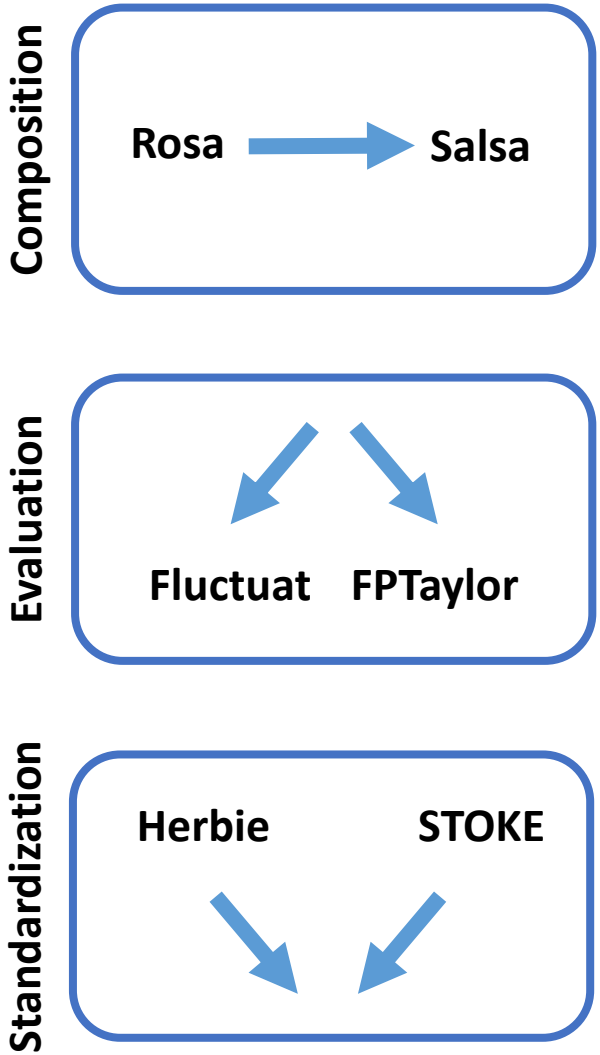
Rapid improvement in hard problems!

We want our community
to keep progressing!

As community grows,
growing pains appear

# Growing pains

Similar growing pains in compilers, HPC, SAT, SMT, … communities




**Rosa:** `def example(x: Double): Double = …`

$\sqrt{x+1} - \sqrt{x}$

**Salsa:** `double example(double x) { … }`

**Fluctuat:** `Poly, Inv, F1a, F1b, idem, …`

**FPTaylor:** `sine, sqrt, verhulst, …`

**Herbie:** `ulp(NaN, Inf) = UINT_MAX`

**STOKE:** `ulp(NaN, Inf) < UINT_MAX`



# FPBench

FPBench is **community infrastructure**
for **cooperation** and **comparison**
in the FP community.

http://fpbench.org




# FPBench

β

Common format

Benchmark suite

Named measures




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$$\sqrt{x+1} - \sqrt{x}$$

Arguments

```
(FPCore (x)
```

```
 (- (sqrt (+ x 1)) (sqrt x)))
```

S-expression syntax

$$\sqrt{x+1} - \sqrt{x}$$

```
(FPCore (x)
  :name “Sqrt Difference”
  :cite (hamming-87)
  :pre (> x 0)
  (- (sqrt (+ x 1)) (sqrt x)))
```

Metadata

Preconditions

```
(FPCore (x0)
  :name “Sine Newton”
  :cite (darulova-kuncak-2014)
  :pre (< (abs x0) 1)
  (while (< i 10)
    ([i 0 (+ i 1)]
     [x x0 (+ (+ (- x (/ (pow x 3) 6.0))
                 (/ (pow x 5) 120.0))
                 (/ (pow x 7) 5040.0))
              (+ (+ (- 1.0 (/ (* x x) 2.0))
                 (/ (pow x 4) 24.0))
                 (/ (pow x 6) 720.0))))])
    x))
```

Loops

Common functions

# FPCore common format

Simple to use

S-expression syntax
Purely functional
No control flow analysis

Expressive

All C, Fortran functions
Loops, conditionals
Tools support parts

Extensible

Metadata properties
Tool-specific metadata
Input or output format



Generate from higher-level, imperative FPImp lang.



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Simple to implement

Covers all existing uses

Simple to extend, specialize



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# FPBench benchmark suite

72 total benchmarks

Drawn from existing papers

Annotated with source, ranges, description, citation

# FPBench benchmark suite

| Existing programs | | Rich features | | Diverse domains | |
|---|---|---|---|---|---|
| FPTaylor | 29 | Arith | 72 | Textbook | 59 |
| Herbie | 28 | Expt | 16 | Math Alg | 6 |
| Rosa | 6 | Trig | 11 | Emb Sys | 4 |
| Salsa | 9 | Loop | 12 | Sci Comp | 3 |
| | | Branch | 3 | | |



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## Common format

Simple to implement

Covers all existing uses

Simple to extend, specialize

## Benchmark suite

From existing projects

Cover many domains

Grows over time

## Named measures



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# FPBench measures

Formal definitions of accuracy measures

Described along 5 axes

Standard measures so tools agree

# FPBench axes of measurement

Scaling vs. non-scaling

Absolute, relative, ULPs, bits, …

Forward vs. backward

Fixed input error vs fixed output error

Maximum vs. average




Sound vs. statistical

Formal guarantees vs mathematical accuracy

Improvement





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Cover many domains

Grows over time

## Named measures

Terms for measuring error

Standard across tools

Flexible but rigorous



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